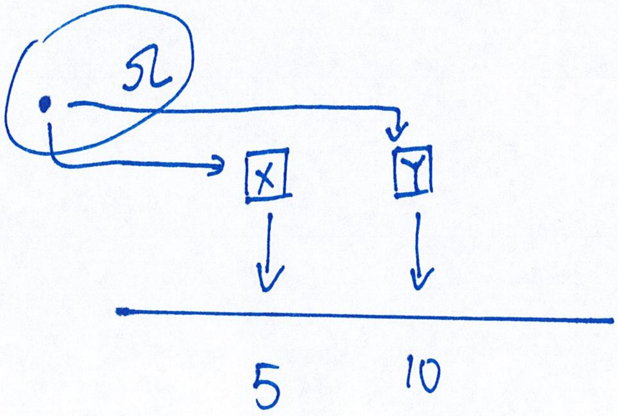
Joint probability mass function

Def. Consider a random pair (X, Y).

Its joint PMF is

$$P_{X,Y}(x, y) = P(X=x,\ Y=y)$$

$$= P(\{\omega \in \Omega \mid X(\omega)=x,\ Y(\omega)=y\})$$



$y \backslash x$	1	2	3	4
4	$1/20$	$2/20$	$2/20$	
3	$2/20$	$4/20$	$1/20$	$2/20$
2		$1/20$	$3/20$	$1/20$
1		$1/20$		

Consider a discrete random pair (X, Y) with joint PMF displayed in the table above.

- $P(X \leq 2, Y \leq 3) = \frac{2}{20} + \frac{4}{20} + \frac{1}{20} + \frac{1}{20}$

$$= \frac{8}{20}$$

- $P_X(2) = P(X=2) = \frac{2}{20} + \frac{4}{20} + \frac{1}{20} + \frac{1}{20}$

$$= \frac{8}{20}$$

▣ Properties of joint PMF / CDF.

- $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$

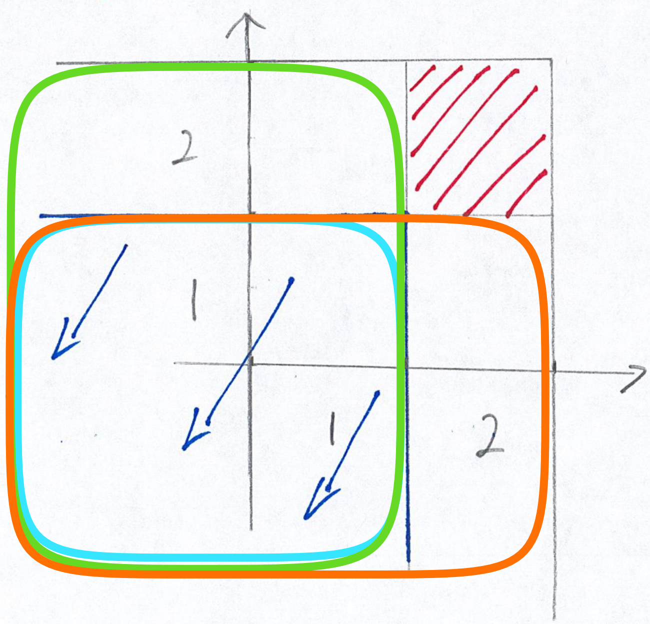
$$= \sum_{\forall (k_1, k_2) \leq (x, y)} \sum P_{X,Y}(k_1, k_2)$$

- $\underbrace{P_X(x)}_{\text{marginal PMF}} = \sum_{\forall y} P_{X,Y}(x, y)$

- $\sum_{\forall x} \sum_{\forall y} P_{X,Y}(x, y) = 1$

- $P_{X,Y}(x, y) = F_{X,Y}(x, y) - F_{X,Y}(x^-, y)$

$$- F_{X,Y}(x, y^-) + F_{X,Y}(x^-, y^-)$$



1 " = " 2 −

NOTE: If $P_{X,Y}$ is given, then we can find P_X and P_Y.

However, knowing P_X and P_Y is NOT sufficient to construct $P_{X,Y}$.